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2. Nano Materials

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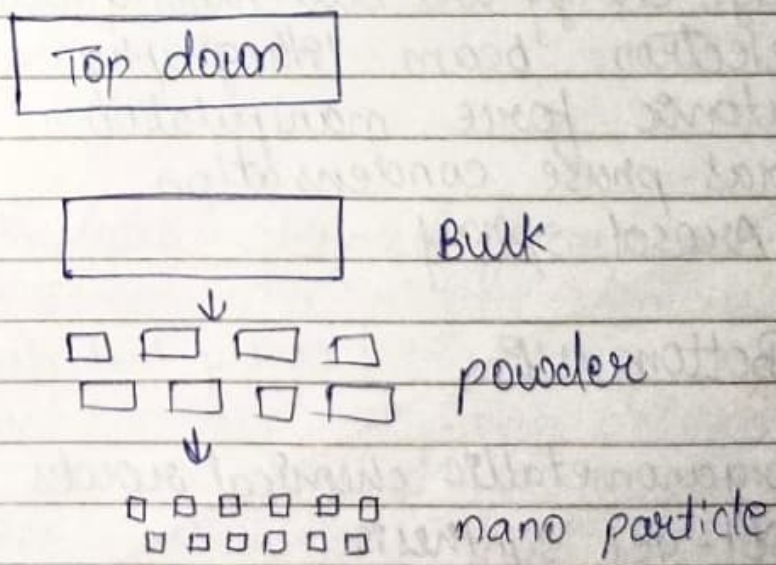
Nano material ⇒ Anything which on the scale is less than 100nm.

		Macro study	Nano study
a)	copper	opaque	transparent
b)	Platinum	Inert	catalyst
c)	Aluminium	stable	combustible
d)	Silica	Insulator	conductor
e)	Gold	solid	liquid
		Inert	catalyst
		Yellow	Red

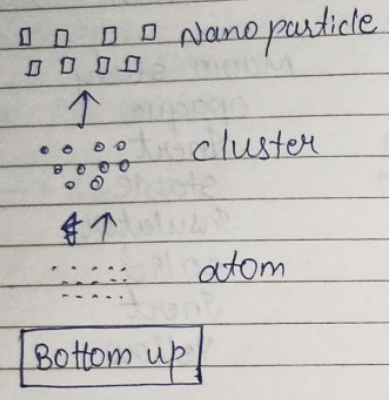
Materials which show its property in macro state and suddenly changes in nano state. It is considered as nano materials when 75% or more shows ~~the~~ its property of nano particle

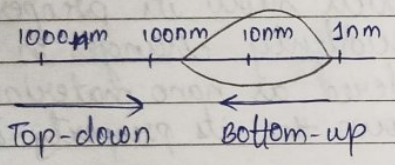
* Methods of synthesis

a) Top-down approach $10^{-9}m$ = size



b) Bottom-up approach





Process

a) Top down [easy but generates waste & dirtop]

i) High energy wet ball milling
ii) electron beam lithography
iii) atomic force manipulation
iv) Gas-phase condensation
v) Aerosol spray

b) Bottom-up [costly but doesn't generate waste]

i) organometallic chemical routes
ii) Sol-gel synthesis

* Carbon NanoTubes (CNT)

CNT are hollow composed of c-atom, arranged in hexagonal lattice.
Typically measured in 1-3nm in diameter

- Properties

1) More hardness as compared to diamond & strength 10 times more than steel.
2) low gravitational weight. 1/4th dense as steel.
3) High heat capacity, making them resistant to expansion when heated.
4) Each c-atom form covalent bond with three others, creating a hexagonal lattice.
5) chemically inert and corrosion resistant.

* Types of Carbon NanoTubes (CNT)

1) Single walled CNT [SWCNT]
2) Multi walled CNT [MWCNT]

1] SWCNT

i) It is cylindrical, formed from ~~roo~~ rolled graphen sheet, with one-dimensional shape
ii) Are armchair and zig-zag determined
iii) Armchair single walled CNT are conductive, others are semiconductor.

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iv) Unique electronic properties are due to their specific lattice arrangement.

v) Exceptional Strength ~~ff~~ faribility (flexibility) and electrical conductivity makes valuable in various technological (technical) application.

- Properties SWCNT

i) Act as semiconductor
ii) Band gap range from 0 to 2eV
iii) Are used in circuit dissassembly due to their specific electrical properties.
iv) Shape - tiny
diameter ⇒ 1-2nm
length ⇒ 1-2 um

- Application

i) ~~SI~~ SWCNT serves as multi functional additives enhancing diverse material property at ~~minimal~~ minimum concentration. Industry utilise them in composite, coating, tyres, structural component and electro chemical device.
Secondary application include adhesive, lubricants, antistatic products, conductive film and cable etc.

i) In composite material:
SWCNT are finding their ways into industry like alternative energy, automotive and aerospace. They are used to strengthen material (composite) in wind turbine blade, boat hulls and aircraft part making them lighter and stronger

ii) In Rainforced Plastic:
SWCNT are used in automotive and air space. In automotive they help reduce the weight of car components which is crucial for improving fuel efficiency and ~~load of~~ reduce the load on batteries.
In Aerospace these material needs to withstand at extreme condition and load. CNT offers them stronger material and durability is increased

iii) In Industrial Coating:
SWCNT are added to primers and paints for vehical coating, equipment and storage container due to its durability and anti-static properties.

iv) In car tyres:
CNT boost tyres quality by strengthing stiffness, tear resistant, fuel efficiency and grip, requiring only small amount and no change to production method. They improve electrical conductivity without extra

fillers like carbon black. SWCNT are expeded to be used in 30% of premium tyres and 10% of standard tyres

v) In electrochemical Power sources:
- Used in rechargable batteries for electric vehicles, laptops and mobile phone because it boost storage capacity and life span
- Lithium-ion batteries and alkaline cell

vi) Enhancing Rubber Goods:
strength elasticity, absorption resistance and colour retention it covers the 1/3rd portion of rubber industries.

2] Multi walled carbon Nanotube [MWCNT]

MWCNT's are like scrolled up scrolls made of several layers. They have both inner and outer diameter. It have good storage and more resistant to chemical reaction.

- Properties

i) Outer diameter (2-20 nm)
ii) Inner diameter (1-3 nm)
iii) Length (5-6 um)
iv) Exhibit good tensile strength due to strong sp bond between carbon atoms

v) When added to material like thermoset or thermoplastic, greatly increases its strength and storage capacity.
vi) Exhibite outstanding chemical and thermal stabilities, enduring the temperature exceeding 600°C.

- Applications

i) In Batteries:
Replacing graphite with MWCNT doubles the storage capacity of lithium-ion batteries which is essential for portable electronics.
MWCNT based electrode are lighter, thinner and 10 times more conductive then amorphous carbon reducing battery weight and power need especially in electronic vehicle.
In super capacitor MWCNT increases power density enabling faster charging time for the device like cellphone and laptop.

ii) In solar cell:
Solar cell with tall tower of MWCNT grown on iron coated silicon sheet absorb more light. Each tower contain millions of vertically aligned MWCNT leading to efficiency peak at 45°-90° to the sun. making them ideal for space application without requiring mechanical orientation.

iii) In transistor:
MWCNT ~~play~~ can be used as transistor in modern integrated circuit offering a potential alternative to traditional silicon switches. They may initially require very low temperature and operate at the scale of individual electrons.

iv) In nano Electronics:
MWCNT play a vital role in nano electronics due to ~~to~~ their high conductivity offering potential as semiconductor alternative to the silicon with conductivity influenced by their structure.

v) In enhancing Flat Panel display:
MWCNT are used in flat panel display forming cathode strip on glass plate and interacting with phosphorus coated Indium Tin oxide (ITO) anode strips to create pixels.

vi) In energy production and storage applications:
MWCNT are essential for energy production and storage offering superior property for electrodes in batteries fuel cell and electrochemical application compared to traditional carbonecious material like carbon fiber and graphite.

* Applications of Nanomaterial

1. In electronics:
Electronics are crucial in our daily lives powering device like Smartphone, computer and TV nanomaterials such as carbon nanotubes and graphen are revolutionery material for electronics by offering better creation of thinner, lighter and more efficient electronic circuit and display.

Nano electronics focus on making electronic device smaller, lighter and more powerful by manipulating CNT material at the nanoscale, scientist can unlock new branch for electronics. This field hold promise for ~~ampo~~ improving every efficiency ~~of~~ enhancing device performance for wearable technology. One significant application for nanoelectronics is in energy storage and energy generation. Nanomaterial enable the development of display screen with lower power consumption and memory chip capable of storing vast amount ~~of~~ in compact space. Wearable electronics such as smart watches and fitness tracker benefit from nanomaterial, light weighted and energy efficient properties. moreover, nano electronics contribute to advancement in green energy technology like solar cell and super capacitor offering more

Sustainable & efficient power solution. Moreover or we can say over all nano material play a vital role in ~~saving~~ safing the future of electronics & making the way for smaller, smarter & more ~~effict~~ energy efficient devices.

2. Uses of Nanomaterials in sensors :
Sensors are vital tools for detecting physical, chemical or biological signals can be greatly ~~the~~ enhanced by incorporating nano materials. These tiny structure sensors by boosting sensitivity, selectivity and speed of response. For ex : nanoscale particles and films ~~im~~ empower sensors to detect ~~ve~~ even small quantity of gases toxin or bio-molecule. These sensors can seamlessly integrated into wearable or implantable device, simplyfying health monitor moreover nanomaterials unlock novel sensors like nano photonic sensors which exploit light to detect changes in refractive index or nano mechanical sensor which utilize vibrations to sense mass or force alteration. These advancement marks a significant state in sensor technology offering more accuracy, more efficiency and more adoptibility across numerous application.
From monitoring air quality and detecting pollutants to tracking vital sign in healthcare nano material based sensors promise good impact. They enable us to better understanding and interact with our environment addressing challenges more ~~effec~~ effectively as nano technology research progress the potential for further innovation in sensor design and functionality grows exponentially. Nanomaterial enhance sensor represent a corner stone of modern technology leading to reset industries with their ~~unparallel~~ unparallel capabilities. These sensors make a future where our ability to detect & response to diverse stimuli is more precise, efficient & accessible than ever before.

3. Application of nanomaterials as catalysis :

a) ~~Hydrogen storage~~

Nanomaterials are revolutionizing catalysis by enhancing how catalyst with substance. They are designed to react with their tiny size and unique surface property make them highly effective in various catalytic process

a) Hydrogen storage
→ CNT are tiny tube made of carbon atom scientist are interested in them because

they can store hydrogen gas, research has shown that CNT can hold more hydrogen than other materials even at room temperature. This makes them promising for storing hydrogen which is a clean & efficient energy source.

b] Water Purification

→ Nano particles made of Titanium dioxide (TiO_2) are used to clean water. TiO_2 is a special material that can absorb light and create a reaction that can clean the water. When light heats TiO_2 nano particle in water it makes electrons move around which helps to breakdown harmful substance in water. Scientist are also looking into using silver nano particle to make water filter that can kill bacteria and keep water clean.

c] Carbon Monoxide oxidation in catalytic converter

→ Catalytic converter are device in car that clean up harmful gases from exhausted gases. Nano materials are tiny particles measured in nm (nanometer) are used in catalytic converter to clean up CO emission. This catalytic converter is more efficient at reducing pollution from car.

* Uses of Nanoparticle Nano material in environment

1. Solar cell
2. Smartphone Displays
3. Sensors
4. Electronic Devices
5. In batteries
6. water purification
7. In transistors
8. In Reinforced Plastics
9. Used as catalyst

a] As a cleaning pollution
b] Purifying water
c] Filtering air
d] Fixing soil
e] Sensors for monitoring
f] Generating clean energy
g] Managing waste
h] Monitoring the environment.
i] Restoring the Ecosystem.

Nano material has vast application or uses in environment such as cleaning

a] As a cleaning pollution

→ Nanomaterial like graphene oxide can absorb oil in water and Titanium di oxidant nano particles break down pollutants in soil

b] Purifying water
→ Nano particles such as CNT are used in water filter to remove contamination and silver nano particle can kill the water in water purification system.

c] Filtering Air
→ Air purifier with nano material filter like nano fibers made from polymer or metal oxide. It can trap harmful partiile & gases & can improve air quality

d] Fixing soil
→ Nanoparticles like iron oxide can bind the heavy metal in soil making them easier to remove while nano scale iron can degradate organic pollutants

e] Sensors for monitoring
→ Nano particle based sensors such as CNT sensor can detect volatile organic compound & help to provide real time data on environmental parameter.

f] Generating clean energy
→ Quantum dot enhance the efficiency of solar panel by converting more sunlight into electricity & nano catalyst improve the performance of fuel cell for cleaning the energy production.

g] Managing waste
→ Nano material like nano zero violent iron breakdown hazardous waste into harmless by product

h] Monitoring the environment
→ Smart nano sensors embedded in environmental monitoring device like wearable air quality monitor can track ~~air~~ pollution level & provide alert to the user.

i] Restoring the Ecosystem
→ Nanoparticles like bio-char enhance soil fertility by improving water absorption & nutrient. It can promote plant growth & ecosystem restoration.

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